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Completion and Multi-Stage Infill Frac Methodology to Unlocks the Potential of Tight Oil Reservoir

Yong Ouyang, Petroelum Development of Oman, Oman; Ishmeev Timur, Abraj Energy Services, Muscat, Oman; Al Ghammari Ahmed, Johnny Falla, and Sayapov Ernest, Petroelum Development of Oman, Oman

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Abstract

An oil field in Southern Oman, producing from a tight, sour salicylate reservoir, faces a steep production decline after 24 years on stream. Maximizing recovery from the 300-meter pay requires infill selective slickwater stimulation. However, isolating existing perforations using mechanical completions is cost-prohibitive and operationally complex, limiting high-rate treatments. This paper presents a successful chemical diversion fracturing strategy, enabling controlled fracture propagation across closely spaced zones, combined with a high-viscosity friction reducer (HVFR) based on high salinity water trial to optimize placement, efficiency and ultimately to keep the project within the initial budget.

A gas injector was treated in November 2024. The pay zone at 4,100 mMD had four existing perforations intervals below. Four-stage fracturing stimulated the top 80 m pay zone. Completion strategy on Stage 2 (perforations at 4,067-4,070 m and 4,056-4,059 m), where chemical diversion technique was employed to place two sequential fracture treatments (2a and 2b). Stage 2a was pumped with 54 metric tons (MT) of proppant, followed by pumped chemical diverter, and then Stage 2b with 76 MT of proppant. High-rate pumping of up to 8 m³/min tested the limits of the diverter's effectiveness, maximizing the stimulated rock volume and assessing its ability to maintain zonal isolation. DFIT and radioactive tracer were conducted to verify geomechanics and diversion efficiency. Accessible fresh water logistic related cost can reach six digits but a new HVFR frac fluid tolerant to high salinity water deployed on Stages 2a, 2b and 3 enhanced operation efficiency and reduced total operating cost.

Pressure trends and post-treatment tracer logs confirmed successful diversion. With Stage 2a fractures propagating through both 2a and 2b perforations, while Stage 2b fractures exhibited clear isolation from the lower section. These results validated the initial breakdown test data, highlighting the chemical diverter efficiency. Selectively placed fractures in a high-stress, low- permeability laminated formation met the objective. The new HVFR frac fluid demonstrated successful application with high-salinity water, proving its viability as an alternative to fresh water. Total operating cost was reduced by 25% with application of chemical diverter and New HVFR fluid system.

Keywords: Infill Frac, Chemical Diverting, HVFR, High Salinity Water, Tracer Log, Stress Contrast

Introduction

The AT Formation in Oman represents one of the most technically challenging and operationally constrained tight oil reservoirs within the Southern Oman Basin. The formation is a laminated silicilyte rock with low permeability and high sour oil content. After 24 years of production and depletion from earlier completions, infill stimulation becomes essential to access bypassed reserves and sustain reservoir pressure via gas injection. However, conventional mechanical isolation is economically and operationally prohibitive, leaving chemical diversion as the only viable option.

This paper documents the multi-stage hydraulic fracturing methodology implemented on a gas injector well, A-42, focusing on novel HVFR fluid systems compatible with high salinity water and chemical diversion strategies to manage closely spaced perforation clusters and ultimately, the cost of the well completion and workover. The workflow involved real-time geomechanical evaluations, laboratory fluid compatibility testing, high-rate pump trials, detailed fracturing stage- by-stage execution strategies, and tracer-based fracture diagnostics designed to maximize operational efficiency, mitigate fracture propagation risks, and achieve reliable zonal isolation. The results of the technical, operational and logistics efforts unlock broader opportunity in the field to maximize the recovery.

The AL field in Southern Oman is known for its challenging AT silicilyte formation—a tight, laminated, and partially shaly rock encased within thick A salt intervals. The deposit is located in a 3 km deep formation which is primary silica, is a unique source and reservoir rock. The rock is organic rich, finely layered, has high porosity (Up to 30%) but low permeability (~ 0.01 md). Net pay thickness varies from approximately 250 m in flank areas to 350 m in the crest.

Field and Reservoir Geology

The AT silicilyte reservoir is part of the Neoproterozoic–Cambrian Huqf Supergroup, lying at depths around 4,000–4,200 m in the South Oman Salt Basin. The formation is a silica-rich, laminated, and partially shaly mudstone, sometimes locally referred to as "silicilyte" due to its unusually high silica content, low clay fraction, and brittle mechanical behavior. Despite its fine lamination and tight permeability (~ 0.01 mD), the rock exhibits high porosity (up to 25%) and moderate total organic content, suggesting that it functions as both a source and reservoir rock.

The reservoir is encased above and below by thick A salt sequences—part of the regionally extensive evaporite cycles (A A–E) of the Huqf Supergroup. These salt layers act as both vertical seals and fracture barriers, playing a key role in pressure retention, fracture height containment, and long-term hydrocarbon entrapment.

In many ways, the AT silicilyte shares geological parallels with other globally recognized tight, siliceous, source-reservoir systems such as:

- The Barnett Shale (USA)
- The Vaca Muerta (Argentina)
- The Qusaiba Shale (Saudi Arabia)

However, its combination of deep burial, high porosity, and laminated silica-rich texture makes it a distinct target within the Middle East context—offering both challenges and opportunities for advanced hydraulic stimulation strategies.

Evolution of Hydraulic Fracturing Strategy

In the past 20 years, more than 40 wells in the AL field were hydraulically fractured using guar-based polymer fluid systems with intermediate-strength ceramic proppant (ISP). These earlier stimulation campaigns typically applied two to four fracture stages across the ~ 300 -meter pay zone. While technically

consistent with regional practices of the time, these designs failed to fully access the reservoir volume due to both economic and technical constraints.

Contributing factors included:

- Constrained budgets that limited stage count and fluid selection
- Overestimation of fracture height growth, leading to large uncontacted sections
- Lack of diagnostic tools to verify fracture coverage or zonal efficiency

Compounding this underutilized reservoir pay (literally: under-stimulation issue), the original completion strategies introduced a new challenge: many wells retained open perforations from prior treatments that now serve as dominant fluid loss pathways, complicating access to unstimulated pay in between. Creating new perforation clusters risks leak-off or premature fracture reactivation through old intervals, making effective zonal isolation a critical and previously unresolved constraint.

Due to the low KV/KH ratio (~ 0.01), hydrocarbon flow was negligible from unfractured intervals, as confirmed by MPLT and SNL logging. The limitations of previous designs resulted in an overall recovery factor of less than 10%.

Legacy fracture treatments in the AL field primarily used guar-based polymer fluids, with proppant placement and slurry rates limited to 3.5–4.0 m³/min. These early designs typically yielded estimated fracture half-lengths of 40–70 meters, as interpreted from pressure-transient analyses and diagnostic well tests.

In subsequent trials, the field shifted toward HVFR-based systems, which enabled higher slurry rates up to 8.0 m³/min. This increase in pump rate corresponded to well test interpretations suggesting fracture half-lengths of up to 130 meters. While such high-rate slickwater treatments often result in complex or non-planar fracture geometries, the modeling and test results were accepted as a baseline for ongoing design optimization—including the infill fracturing campaign at A-42.

The newly adopted infill fracturing strategy, as demonstrated in the A-42 campaign, marks a significant departure from legacy practices. It emphasizes selective infill fracturing of bypassed zones in existing wells, rather than drilling and stimulating new ones—achieving potential cost savings of up to 60%. Based on enhanced understanding of fracture geometry and cluster interaction, a stage-wise completion design may apply, targeting 25–30 m net pay per stage, with an average of 8–12 stages per well. This configuration requires approximately 6,000–9,000 m³ of water per well.

Logistical constraints further shaped the evolution of this approach. Historical records from the field show that non-productive time (NPT) due to freshwater unavailability accounted for 5–10% of total well time. This challenge, combined with the remote location of the AL field, underscored the need for alternative water sources. High-salinity water from the A field (available at 5,000 m³/day) was identified as a viable option, leading to the development and deployment of HVFR systems compatible with salinity levels reaching 30,000 ppm TDS.

This integrated strategy—combining infill completions, chemical diversion, and high-salinity- tolerant fluids—was piloted in A-42 and forms the foundation for scalable recovery enhancement across similar wells in the field.

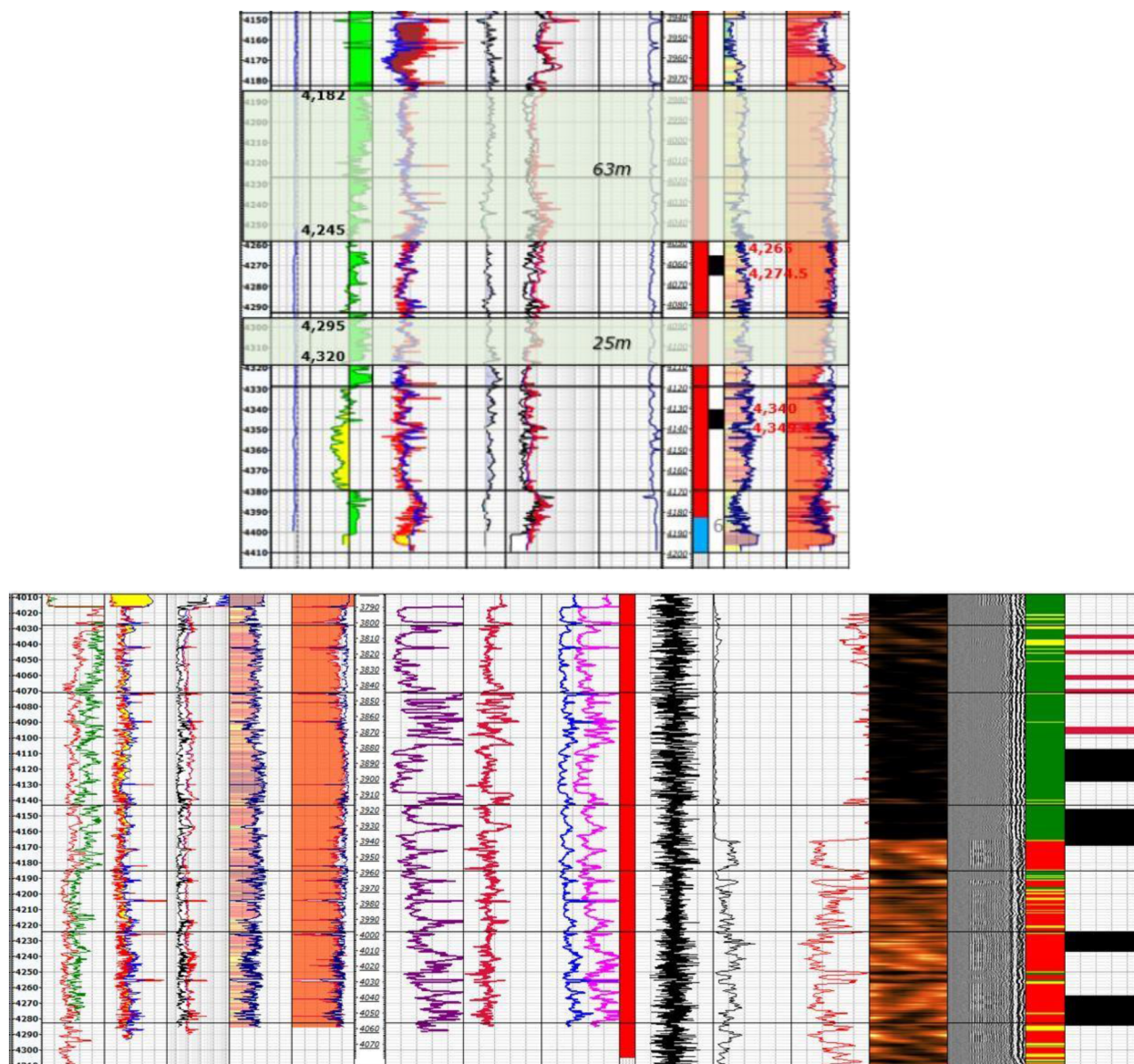


Figure 1—Old Frac Design VS Infill Frac on A-42. The schematic highlights previously placed perforation intervals (shown in black) and newly placed project-specific perforations for the infill fracturing campaign (shown in red).

Project Challenges

Despite the significant potential for enhanced recovery through infill fracturing, the campaign was deferred until recently due to two primary technical challenges:

Zonal Isolation of Previously Stimulated Intervals

One of the most significant technical barriers to successful infill stimulation in the AL field is the presence of existing perforated and stimulated intervals from earlier fracturing campaigns. These legacy zones often remain highly conductive and act as preferential leak-off paths, severely limiting the effectiveness of new fracture initiation in adjacent bypassed zones.

Crucially, these intervals could not be permanently abandoned—they remain strategically important for future injection or production plans. The project therefore required a temporary, non-mechanical isolation solution that would maintain integrity during treatment yet fully degrade afterward.

Traditional mechanical methods such as casing patches or straddle packers were ruled out due to their high cost, operational complexity, and permanent restrictions on wellbore access.

Instead, the project focused on an in-situ degradable diversion method that could:

- Sustain differential pressure during high-rate fracturing
- Provide reliable near-wellbore isolation between tightly spaced clusters
- Fully degrade under reservoir conditions to restore injectivity or flowback

A polylactide-based chemical diverter system was selected for its degradability, sealing capacity, and field deployability under the laminated, high-stress conditions typical of the AT silicilite reservoir.

Water Source Limitation setting High Salinity HVFR requirements.

A second major constraint was the availability of suitable water for stimulation. Freshwater supply via trucking has historically contributed to 5–10% non-productive time (NPT) due to transport delays and logistical bottlenecks.

To eliminate this dependency, the project utilized Amal field high-TDS water (up to 5,000 m³/day), despite its salinity reaching 30,000 ppm—well beyond the tolerance of conventional polymer and basic HVFR fluids.

This necessitated the development and qualification of a high-viscosity friction reducer (HVFR) capable of maintaining:

- Rheological stability and proppant transport at elevated salinity
- Friction reduction suitable for 8 m³/min high-rate pumping
- Clean degradation and low residue under field conditions

The selected HVFR-C system proved effective in laboratory and field trials, enabling full-scale use of local high-salinity water while maintaining operational performance and cost efficiency across multiple stimulation stages.

Project Solutions and Planning

In preparation for the A-42 infill stimulation, several zonal isolation options were reviewed and discussed in multidisciplinary sessions involving completion, stimulation, and geomechanics stakeholders. These conversations considered the operational complexity, cost implications, and geological uncertainties specific to the AT formation—particularly its laminated texture, stress heterogeneity, and existing perforation architecture.

The focus was on identifying an approach that could:

- Enable selective fracture placement between closely spaced clusters
- Maintain operational flexibility potentially avoiding well intervention
- Provide temporary but reliable isolation during high-rate treatments
- Avoid permanent loss of access to legacy perforated zones

Given these requirements, chemical diversion was selected as the base case, offering a balance of simplicity, reversibility, and cost-effectiveness under the prevailing reservoir and operational constraints. A comparative summary of evaluated isolation options is presented in [Table 1](#).

Table 1—Zonal Isolation Method for Infill Frac

Method	HSE Impact	Technical Maturity	Operational Complexity (relative)	Reliability	Estimated Cost (USD)
Casing Patch	Low	Medium (3-month lead time)	High (requires rig/hoist; non-retrievable; reduces tubing ID)	High	\$3–5 million
Straddle Packer	Low	High (readily available)	High (requires rig; intervention required as it is retrievable; reduces wellbore ID)	Moderate	~\$1 million
Cement Squeeze	Low	High	Moderate (requires pre-frac pumping; chemical dissolution for resumption)	Moderate	~\$500K
Chemical Diversion	Low	High (mature and widely applied)	Low (performed during main fracturing treatment; reversible post-flowback)	High	~\$200K

Based on the comparative analysis, chemical diversion was confirmed as the practical and cost-effective isolation method for the A-42 trial. The well was selected to validate this strategy under real field conditions—targeting the top 90 meters of net pay using four fracturing treatments.

Following this trade-off analysis, chemical diversion was selected as the base case for infill fracturing. Well A-42 was chosen for field trial implementation. Four fracturing stages were planned to treat the top 90 m of net pay. Of these, Stage 2 was executed as a split treatment using chemical diversion between two sub-intervals (2a and 2b), while Stage 3 followed as a conventional treatment. The operations incorporated a high-salinity HVFR system and high-rate pumping (up to 8 m³/min) to evaluate both treatment placement and diversion integrity.

Tracer logging and pressure diagnostics were deployed to assess fracture geometry and confirm zonal isolation, providing essential validation of both the design assumptions and field performance.

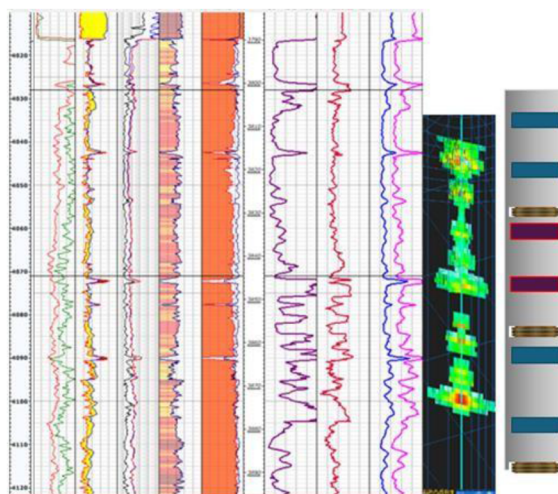


Figure 2—Stratigraphic Column of AT Formation and Pay Intervals from the Discussed Well. The figure presents the vertical distribution of net pay and the planned perforation clusters for the A-42 stimulation. Fracture shapes shown are derived from the fracture design model and included for context—intended to illustrate the anticipated fracture height and stage coverage based on pre-job simulation. This composite view was developed to aid in visualizing the relationship between reservoir layering, completion intervals, and modeled fracture geometry.

Well Completion and Perforation Strategy

Well A-42H1 is a vertical well, completed with a 4.5" monobore tubing and packer. The well was recompleted for infill fracturing with six perforation clusters distributed over three main treatment stages:

- **Stage 1:** 4,081-4,084 m MD & 4,095-4,098 m MD
- **Stage 2a:** 4,067-4,070 m MD
- **Stage 2b:** 4,056-4,059 m MD
- **Stage 3:** 4,034-4,037 m MD & 4,044-4,047 m MD

Extensive internal discussions were held during planning to determine optimal perforation cluster placement. The final placement prioritized the interval believed to have lower in-situ stress, as supported by early geomechanical interpretations. Fracture simulation runs were used to ensure that perforations would be effectively contacted by the expected fracture geometry. The planning objective was to initiate in the zone most likely to take fluid, then assess redirection feasibility through diversion. To enhance the 3-D reservoir and fracturing simulation the GOHFER modeling was used to optimize cluster spacing and anticipate fracture height growth. Particular attention was given to Stage 2, where the 7-meter vertical separation between clusters introduced elevated risk of cross-propagation during high-rate pumping operations.

Stage 2 was purposefully selected to test sequential stimulation and diversion between two closely spaced intervals. The geomechanical model suggested a difference in minimum in-situ stress between Stage 2a and Stage 2b, implying the potential for controlled diversion using a chemical diverter. See [Figure 3](#) for modeled stress distribution. However, this prediction carried uncertainty, as in-situ stress is not the sole factor governing fracture propagation. The principal stress gradient is a key parameter, but effective stress, rock mechanical heterogeneity, pore pressure distribution, and natural fracture networks also influence the final fracture path. For instance, a lower effective stress does not guarantee dominant fracture growth if near-wellbore tortuosity or anisotropy impedes propagation. The interplay of these subsurface dynamics made Stage 2 a critical pilot for verifying whether real-time diversion could be reliably achieved between two adjacent zones under variable stress and high-rate conditions.

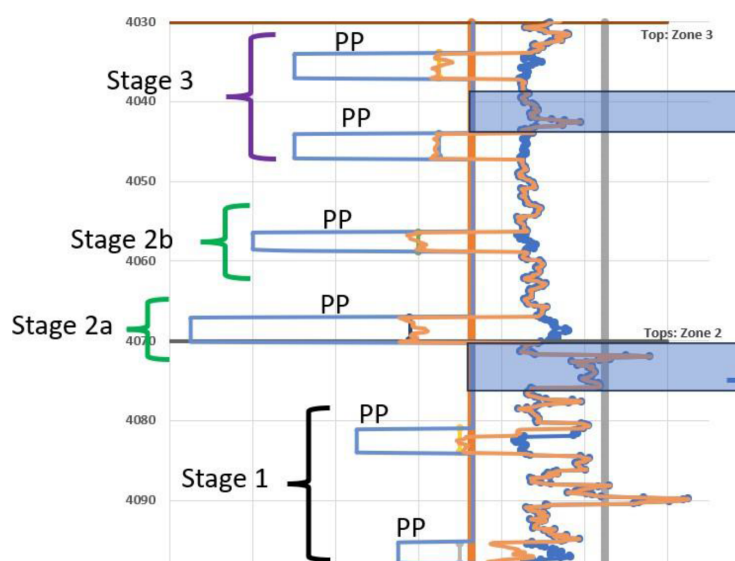


Figure 3—Reservoir Geomechanical Model Interpretation. The model shows the distribution of minimum horizontal stress across pay intervals in the A-42 well, including Stage 2a and 2b. This stress contrast was a primary design input for the diversion strategy.

These pre-job insights informed the staging and spacing of perforation clusters, especially in Stage 2 where the minimal vertical offset and expected stress contrast made it a suitable pilot zone for testing sequential chemical diversion. The combination of geomechanical modeling and targeted perforation design created the foundation for real-time fracture execution, which is detailed in subsequent sections.

Laboratory Qualification: Fracturing Fluid Selection and Diverter Qualification

A critical element of the project involved laboratory qualification of the fracturing fluid system and chemical diverter, especially considering the deployment of high-TDS water from the Amal source. The objective was to identify a friction reducer compatible with high-salinity water, sufficient viscosity, proppant-carrying capacity, and clean-up behavior under high salinity conditions.

Fracturing Fluid Selection

Abraj Energy Services conducted a comprehensive laboratory program evaluating three HVFR candidates:

- **HVFR-A:** Optimized for fresh water
- **HVFR-B:** High salinity tolerance up to 60,000 ppm TDS
- **HVFR-C:** Advanced formula requiring lower dosage at high-TDS

Testing protocols included:

- Rheological performance at varying temperatures and shear rates.
- System should be able to achieve 25 cp at 170 sec⁻¹ at 25° C.
- Oxidizer breaker to reduce viscosity to less than 10 cp at 511 sec⁻¹ at 80° C.

System should provide a minimum fracture retain conductivity of 80%. System with 0.75 l/m³ concentration of HVFR should provide minimum 70% Friction reduction comparing with fresh water, equivalent 3.74" ID pipe at 4 to 8 m³/min rate. All of the above shall exhibit tolerance with high-salinity water.

HVFR-C demonstrated the most favorable profile under Amal water conditions (refer to [Figure 4](#)), maintaining sufficient viscosity and transport properties while minimizing chemical dosage and cleanup complexity. The product also exhibited consistent hydration performance, supporting high-rate operations.

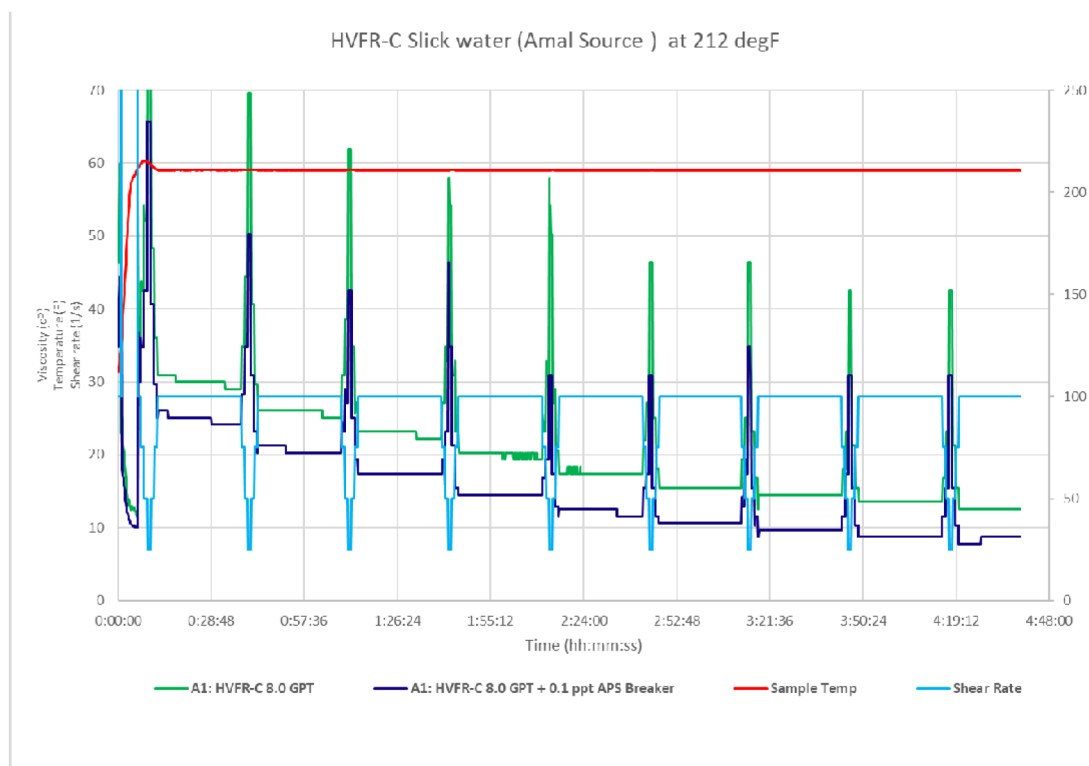


Figure 4—Fluid With HVFR-C Rheology Curves. Graph provides results for baseline and with 0.1 PPT of live breaker

Diverter Qualification and Breakdown Behavior

Parallel testing of chemical diverters evaluated multiple particle size distributions and degradability profiles. Emphasis was placed on matching diverter breakdown time with post-frac flowback schedules to avoid flow impairment while ensuring temporary zonal isolation.

Diverter effectiveness was tested under temperature-controlled environments simulating bottomhole conditions. Results confirmed that the selected diverter provided reliable sealing capacity and clean breakdown, meeting operational requirements for infill fracturing under high- rate, high-salinity conditions.

Selection of the diverter system required careful consideration of both operational robustness and degradation reliability. The chosen diverter, a polylactide-based resin, was evaluated against alternatives that exhibited slower breakdown profiles. The material needed to temporarily withstand differential pressure and elevated temperatures—up to $\sim 92^{\circ}\text{C}$ —without premature failure, yet degrade within an acceptable timeframe to avoid impairing long-term injectivity.

Thermal degradation studies (refer to Figure 5) compared two diverter formulations. One sample remained largely undegraded after prolonged exposure at 92°C , suggesting potential post- stimulation obstruction risk. The selected formulation, in contrast, exhibited approximately 50% mass reduction after 50 hours under the same conditions—balancing mechanical sealing with predictable breakdown.

To ensure field compatibility, crude interaction tests were conducted using samples from offset producers, even though A-42 functions as an injector. These trials confirmed the absence of adverse reactions or agglomeration between the diverter and reservoir hydrocarbons, mitigating cleanup risks during flowback or injection restart.

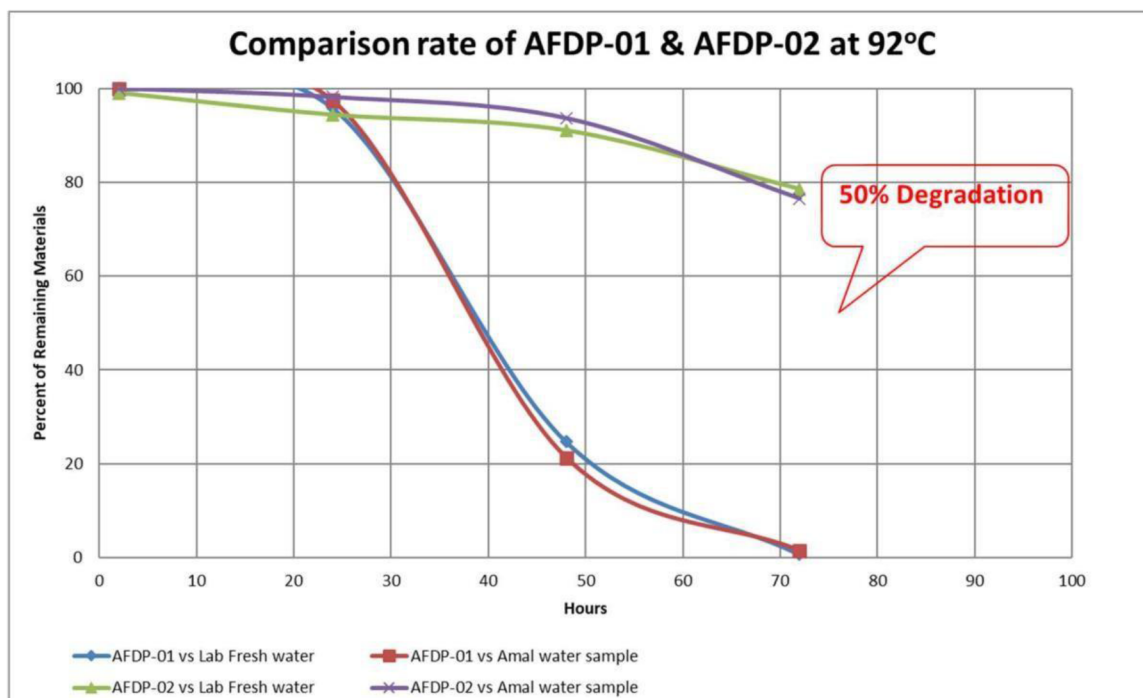


Figure 5—Diverter Performance Based on the Temperature. Results provided option from two types of diverter. Diverter with degradation reaching 50% quicker was selected.

Diversion Strategy and Diagnostic Verification

Effective diversion between closely spaced clusters was central to the A-42 stimulation strategy. In particular, Stage 2 presented a technically challenging scenario: two perforation clusters (2a and 2b) spaced only 7 meters apart vertically, within a laminated formation exhibiting limited natural stress contrast. Traditional mechanical isolation was ruled out due to complexity and well access constraints, placing operational reliance on chemical diversion.

The completion plan adopted a sequential fracturing approach:

- Fracture Stage 2a first
- Deploy a degradable diverter
- Stimulate Stage 2b immediately afterward

This method relied heavily on accurate modeling and real-time fracture response to validate the diversion outcome. Pre-job geomechanical analysis suggested a modest differential in minimum horizontal stress between the two intervals—enough to potentially enable upward redirection of fracture energy if the lower cluster could be effectively sealed.

Given the uncertainty around stress contrast and the absence of mechanical barriers, the team planned to use a combination of pressure diagnostics, diagnostic fracture injection tests (DFITs), and radioactive tracer logging to evaluate the success of the diversion strategy. Pressure response during pumping was expected to show a marked increase between Stage 2a and 2b if the diverter effectively isolated the lower cluster. Similarly, tracer data would help confirm the vertical extent and confinement of fracture propagation in each interval.

The ability to execute this plan and verify its effectiveness in real time was essential to proving the viability of chemical diversion in low-contrast, laminated reservoirs. The corresponding execution results, including diagnostic data and post-fracture validation, are presented in the following section. Importantly, the insights gained from the A-42 diversion strategy now provide a validated design and execution template

for future wells across the AL field. With many legacy wells having bypassed or under-stimulated pay, the demonstrated ability to reliably isolate and fracture individual reservoir intervals—without mechanical intervention—represents a step- change in field development planning. This learning will directly shape how infill wells and re- completions are approached going forward.

Fracture Design Strategy and Execution

The fracture design for A-42 was developed to validate selective infill stimulation across partially depleted reservoir zones with minimal vertical stress contrast. Special attention was given to Stage 2, which was engineered as a real-time pilot of chemical diversion in a laminated, tight formation with closely spaced clusters. The treatment aimed to maximize stimulated reservoir volume (SRV), improve cluster efficiency, and validate zonal isolation through real-time pressure diagnostics and tracer logging.

Key design considerations included:

- Adapting treatment volumes and rates to each stage's unique geomechanical context.
- Incorporating diversion to isolate closely spaced clusters.
- Leveraging HVFR-C fluid system in high-TDS water to maintain proppant transport and friction reduction.

Due to viscosity limitations of the HVFR-C system at high salinity, the achievable proppant concentration was constrained to a maximum of approximately 360 kg/m³. To compensate, high- rate pumping was critical to deliver sufficient proppant mass and maintain effective transport across both stages.

Design Objectives

- Maximize SRV and cluster efficiency within operational constraints by placing 30/50 ISP proppant reaching 360 kg/m³ concentration
- Mitigate cross-cluster fracture propagation.
- Validate geomechanical assumptions via real-time DFIT and pressure trends.
- Implement high-rate Slick Water/HVFR treatments using high TDS water.

Stage 2 Execution: Diagnostic Fracture Testing and Sequential Diversion

The execution of Stage 2 was deliberately designed as a dual-cluster stimulation to test the viability of chemical diversion under uncertain geomechanical conditions. Stage 2a (4067–4070 m MD) was perforated first, allowing an isolated DFIT to characterize closure pressure, fluid efficiency, and leak-off behavior without interference from the adjacent cluster. The test recorded a closure pressure of 72.11 MPa with a relatively low fluid efficiency of 5.7%, suggesting significant leak-off and possible complexity near the wellbore.

At this point, the original plan had been to proceed directly to the main fracturing of Stage 2a, followed by diversion and stimulation of Stage 2b. However, the field team—led by collaboration between Abraj and PDO—identified a critical opportunity: to conduct a second DFIT after perforating Stage 2b (4,056–4,059 m MD), in hopes of resolving whether a difference in minimum in-situ stress existed between the two clusters. If Stage 2b exhibited a lower closure pressure, the team could anticipate natural fracture initiation from the upper interval. If not, the data would support the need for diverter-assisted flow redirection from 2a to 2b as per original plan.

Once Stage 2b was perforated and tested, the DFIT results revealed a closure pressure of 74.64 MPa and a fluid efficiency of 8.17%. While the efficiency was slightly improved, the closure pressure closely matched

that of Stage 2a. Technical review teams concluded that the observed trend likely reflected continued influence from the initially fractured zone (2a), meaning it remained the lower-stress interval. Consequently, the team planned to initiate fracturing in Stage 2a, deploy a chemical diverter, and then stimulate Stage 2b—actively overcoming the stress differential to direct energy upward. Refer to [Figure 7](#) and [Figure 8](#) for details.

Main Fracturing of Stage 2a and 2b

With the DFIT results analyzed, the treatment moved into the main fracture execution. Stage 2a was pumped first, as designed, using treated water with radioactive tracer #2 to monitor fracture propagation. A total of 66.5 metric tons of proppant was placed at an average rate of 6.01 m³/min. Once the proppant stage was complete, a diverter pill (0.018 MT) was pumped to temporarily seal off the 2a perforations, followed by a high-rate flush and matrix-rate over-flush to stabilize the treatment and minimize fallback.

After a monitored shut-in period to allow fracture to close and pinch placed proppant with pressure stabilization, the Stage 2b main frac was executed using the same HVFR-C fluid system and tracer formulation. 87.5 MT of proppant was placed at an average rate of 7.0 m³/min. The successful pressure response confirmed that diversion had occurred as planned, with fluid flow redirected to the upper cluster. Final over-flush and shut-in monitoring were conducted per design, followed by clean-up operations.

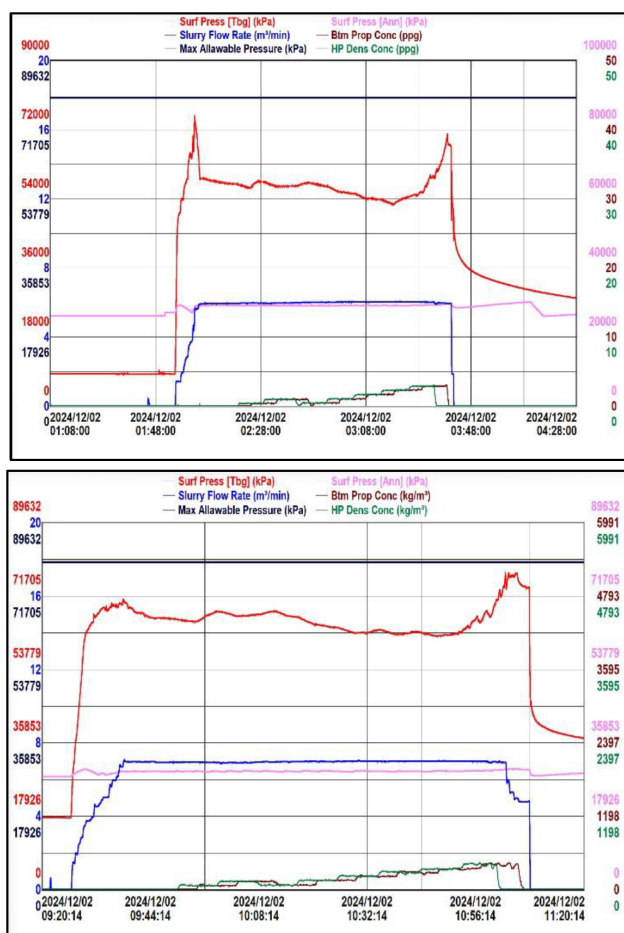


Figure 6—Main Treatment Plots. Top: Stage 2a, Bottom: Stage 2b. Different pressure trend pointed potentially different reservoirs being stimulated and diversion from one zone to another has occurred.

Tracer diagnostics and flowback profiles later validated the sequential fracture placement, confirming the success of both the geomechanical interpretation and the real-time execution strategy.

Stages 2a and 2b Execution Summary Stage 2a:

- Perforated first to preclude preferential propagation into Stage 2b
- DFIT conducted post-perf revealed closure pressure: 72,116 kPa
- Higher leak-off and lower fluid efficiency suggested initial fracture complexity
- Main frac: 66.5 MT proppant, 2.5 PPA max, 6.01 m³/min average rate
- Diverter pill deployed: 0.018 MT

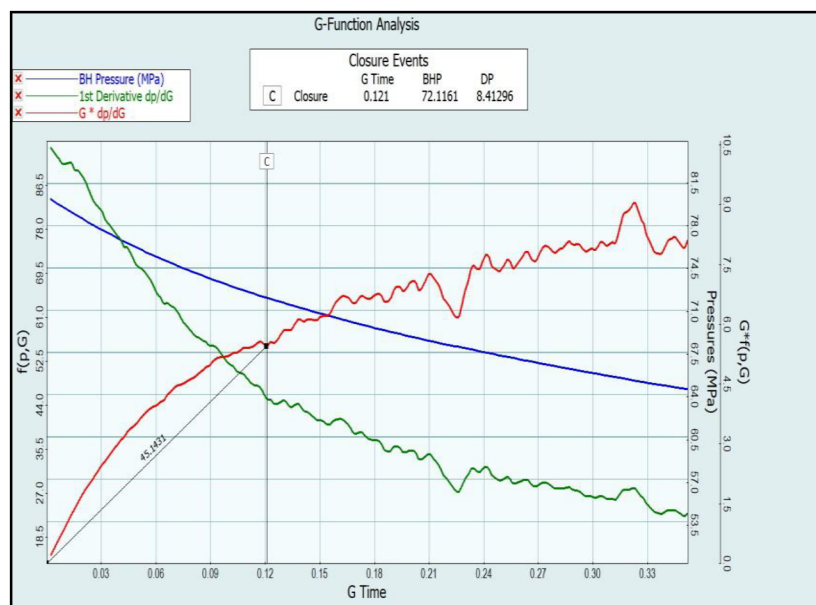


Figure 7—Breakdown Tests and Results from the Zone 2a. The minimum in-situ stress value obtained as 72,116 kPa.

Stage 2b:

- Second DFIT post-perf confirmed comparable closure pressure to Stage 2a: 74,603 kPa
- Fluid efficiency improved, leak-off reduced
- Main frac: 87.5 MT proppant, 3.0 PPA max, 7.0 m³/min average rate
- Diverter effectively redirected fluids, confirmed via pressure trend flattening

These execution results affirmed the foundation for field implementation in Stage 2, where diversion efficiency and fracture confinement were evaluated in real time using sequential diagnostics, tracer logging, and high-rate stimulation.

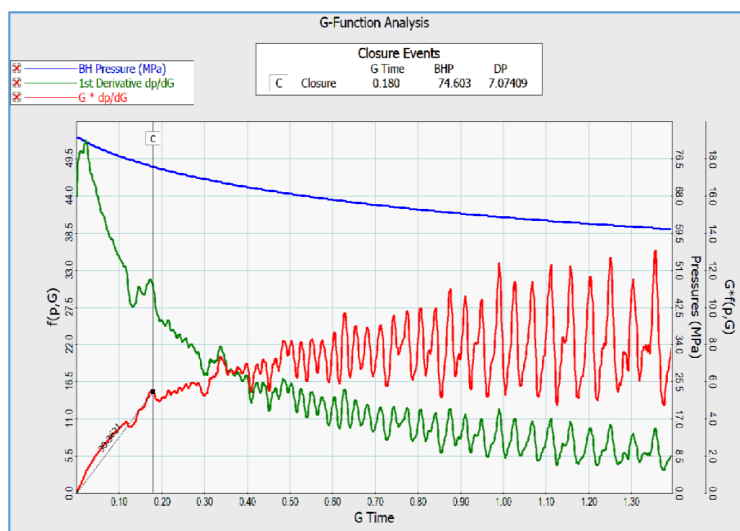


Figure 8—Breakdown tests and results after the second interval in zone 2 was perforated. The minimum in-situ stress value obtained as 74,603 kPa. An additional judgment points at potentially lower value making it corresponding to DFIT during Stage 2a execution.

Fracture Sequential Diversion Diagnostic

The success of Stage 2 hinged on more than just fluid and proppant volumes—it depended on achieving reliable diversion between closely spaced clusters under real-world conditions. The operational plan placed chemical diversion at the heart of the completion strategy, especially given the narrow vertical offset (only 7 meters) between Stage 2a and 2b perforations. With no mechanical isolation in place, and the stress contrast proving subtle, diversion effectiveness needed to be confirmed both through pressure diagnostics and post-treatment logs.

Following the completion of Stage 2a and deployment of the polylactide-based diverter pill, Stage 2b was stimulated immediately. The pressure response during Stage 2b showed a distinct flattening trend compared to 2a, indicating a redirection of flow and successful zonal isolation by the diverter. Moreover, improved fluid efficiency and more stable pressure gradients suggested that the treatment was confined to the upper cluster without loss into the previously stimulated interval.

To further validate the outcome, radioactive tracer diagnostics were deployed across both stages. Post-frac logs confirmed the fracture from Stage 2a had partially extended into the Stage 2b interval prior to diversion. However, after diverter placement, the Stage 2b fracture showed confinement to the intended upper zone. This result aligned with the expected behavior of the diverter under temperature and flow conditions encountered in the field.

The data reinforced two key insights: (1) that carefully timed chemical diversion—supported by tracer and pressure data—can reliably isolate zones in high-stress laminated formations, and (2) that fracture propagation is not solely governed by modeled stress contrast, but by a combination of effective stress, treatment rate, and fluid flow dynamics.

Together, these findings confirmed the value of the chemical diverter system and field execution strategy, establishing a repeatable approach for future infill stimulations in AL and similar fields.

Tracer logs provided definitive confirmation

Post-frac tracer logs showed:

- Stage 2a fractures partially propagated into Stage 2b perforations pre-diverter
- Stage 2b fractures post-diverter confined strictly within the upper cluster

This validated the diverter's performance under high-rate, high-salinity conditions and aligned with pressure monitoring trends.

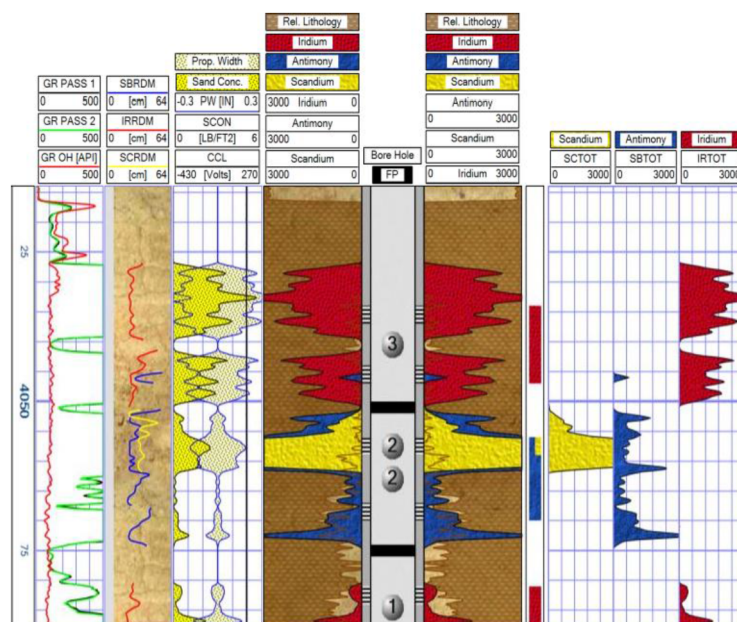


Figure 9—Tracer Log Image Illustrating Fracture Placement before and after Diversion.

The tracer data clearly distinguishes the fracture propagation paths between the two stages. The **blue tracer**, associated with Stage 2a, shows signal response across both 2a and 2b perforated intervals—confirming that in the absence of diversion, the fracture extended through both clusters. In contrast, the **yellow tracer**, injected during Stage 2b, remains confined to the upper perforation interval only. This selective placement highlights the effectiveness of the chemical diverter in achieving zonal isolation and preventing re-entry into the previously stimulated zone.

Operational and Economic Impact

Beyond the technical validation of the stimulation methodology, the A-42 infill frac execution delivered compelling operational and financial returns—primarily due to the strategic use of **high-salinity water** and **chemical diversion**.

Historically, mechanical isolation techniques such as **straddle packers** and **casing patches** have been considered for infill fracturing in the field, but these carry substantial cost and operational penalties. Estimated deployment costs for mechanical systems can range from USD 1 million to over 5 million per well, depending on the scope of intervention and number of stages. In contrast, the chemical diversion system applied in this campaign required no permanent hardware, preserved the full internal diameter of the completion, and was implemented without rig-based operations. In contrast, the chemical diversion system used in this operation—offered **cost savings of over 80%** relative to mechanical alternatives, while also eliminating the need for heavy intervention equipment and preserving completion ID.

In terms of fluid logistics, replacing trucked fresh water with **Amal high-TDS water** brought a further **30% cost reduction** in water handling and storage. The qualified **HVFR-C** system allowed high-rate pumping with efficient proppant transport at reduced polymer loading, which contributed an additional **25% chemical savings**.

Overall, the project realized a **total cost reduction of approximately 25%**, including:

- **Water logistics savings:** ~\$120,000
- **HVFR system chemical optimization:** ~\$80,000
- **Avoided mechanical isolation equipment and intervention:** ~\$800,000+
- **Faster post-frac flowback and minimal deferment losses**

These savings are not just additive—they represent an integrated cost efficiency model that can be scaled across similar infill wells in AL. With growing pressure to extract more from mature assets under tighter budgets, this approach redefines how older completions can be economically revived.

Lessons Learned

The A-42 infill frac campaign yielded valuable insights into the behavior of tight, laminated reservoirs and the execution of non-mechanical zonal isolation:

- **Real-time DFITs are pivotal.** Sequential diagnostics across individual clusters enabled real-time calibration of geomechanical assumptions, allowing data-driven updates to the execution plan.
- **Stress contrast alone is not predictive.** While in-situ stress differences help guide treatment design, actual fracture propagation was also influenced by near-wellbore tortuosity, pressure-dependent leak-off, and local heterogeneity.
- **HVFR performance in high-TDS conditions exceeded expectations.** The selected HVFR-C system-maintained viscosity, friction reduction, and proppant-carrying capacity in high-salinity Amal water, enabling high-rate treatments without the need for fresh water. Its reliable cleanup behavior also minimized chemical residue and flowback risk.
- **Diverter timing and degradability must align with field conditions.** The polylactide-based diverter selected for A-42 successfully balanced isolation performance with clean degradation under reservoir temperature and salinity conditions.
- **Tracer diagnostics provide valuable insight—with limits.** Tracer logs delivered definitive validation of diversion effectiveness but required intervention and deferred analysis. Their use remains valuable but not universally practical.
- **Cross-functional collaboration accelerates decision-making.** Effective alignment between subsurface, fluid chemistry, and operations teams enabled the project to adapt dynamically—such as adding a second DFIT—which was critical to the success of the campaign.

Conclusion

The A-42 case study demonstrates a technically sound, operationally efficient, and economically attractive methodology for infill fracturing in tight, laminated silicilite reservoirs under infrastructure and budget constraints.

The cornerstone of this success was the deployment of a **chemically diverted, high-rate hydraulic fracturing strategy**. The use of a polylactide-based diverter enabled reliable zonal isolation between closely spaced perforation clusters, eliminating the need for mechanical isolation hardware while ensuring controlled fracture placement. This was validated through pressure diagnostics and post-frac tracer logs, confirming effective redirection and confinement of fracture energy.

Supporting this approach, a **high-salinity-tolerant HVFR fluid system** facilitated sustained proppant transport and rate control using local Amal water—removing dependence on fresh water and reducing overall fluid logistics cost.

The combination of these technologies and execution practices led to:

- **Reliable non-mechanical zonal isolation** with effective fracture redirection
- **Controlled and confined stimulation** in laminated pay with minimal cluster interference
- **Over 25% total project cost savings**, driven by chemical diversion, local water utilization, and simplified operations
- **Improved operational flexibility**, including faster flowback and minimal deferment
- This methodology offers a scalable, repeatable solution for maturing fields—one that unlocks bypassed pay without the complexity or cost of re-drilling or mechanical recompletions.
- Looking ahead, the integration of **interventionless acoustic diagnostics** could further enhance real-time decision-making by enabling fracture monitoring without well intervention—amplifying the impact of the stimulation strategy from both technical and economic perspectives.

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Author Contribution

The manuscript was written with contributions from all authors. All authors have approved the final version of the manuscript.

Notes

The authors declare no competing financial interest.

Data Availability

Data will be available based on a request from the corresponding author.

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